

Summations and geometric progressions



Outline

Summations

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Summations



Introduction: Summations

- ▶ In this section we introduce the **summation symbol** $\sum$, a compact notation to write sums of many terms
- ▶ We will see its **main properties** (constants, sum of summations, splitting, index shift and index reflection)
- ▶ Finally, we will compute some **important summations**: the sum of the first n natural numbers, of the first n odd numbers and of the first n even numbers

The summation symbol

Definition 1: summation

Let $a_1, a_2, \dots, a_n$ be n real numbers. Their sum

$$a_1 + a_2 + \cdots + a_n$$

can be written in compact form with the summation symbol:

$$\sum_{k=1}^n a_k$$

which is read: “sum for k from 1 to n of a_k ”. The symbol k is called the summation index.

- ▶ The summation symbol is therefore just a shorthand, which is nevertheless very useful when the terms a_k are defined explicitly as a function of the index k .

Example

$$\sum_{k=1}^{10} \frac{1}{k} = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8} + \frac{1}{9} + \frac{1}{10}$$

$$\sum_{k=3}^n k^2 = 3^2 + 4^2 + 5^2 + \cdots + n^2$$

Dummy index

- ▶ The summation index is a **dummy index**. This means that if we replace k with i , j or any other index (in all its occurrences), the value of the summation does not change.



Example

We have:

$$\sum_{k=1}^n k^2 = \sum_{i=1}^n i^2$$

since both symbols denote the sum of the squares of the first n natural numbers (without zero).

On the other hand, we have:

$$\sum_{k=1}^n k^2 \neq \sum_{k=1}^m k^2$$

since the two symbols denote the sum of, respectively, the first n or the first m squares (if $n \neq m$ the result will be different).

Main properties of summations

We now see the main properties of summations, which allow us to manipulate and simplify expressions involving the symbol Σ .

Product by a constant and constant term

Observation 1

Given $c \in \mathbb{R}$, we have:

$$\sum_{k=1}^n (c \cdot a_k) = c \sum_{k=1}^n a_k \quad (\text{product by a constant})$$

$$\sum_{k=1}^n c = c \cdot n \quad (\text{summation with a constant term})$$

Proof

First property. By the distributive property we have:

$$\underbrace{c a_1 + c a_2 + \cdots + c a_n}_{=\sum_{k=1}^n (c \cdot a_k)} = \underbrace{c (a_1 + a_2 + \cdots + a_n)}_{=c \sum_{k=1}^n a_k}$$

Second property:

$$\underbrace{c + c + \cdots + c}_{=\sum_{k=1}^n c} = c n \quad \text{that is, } c \text{ added } n \text{ times}$$



Sum of two summations

Observation 2

Given two summations $\sum_{k=1}^n a_k$ and $\sum_{k=1}^n b_k$, we have:

$$\sum_{k=1}^n a_k + \sum_{k=1}^n b_k = \sum_{k=1}^n (a_k + b_k)$$

Proof

We have:

$$\begin{aligned} & \underbrace{a_1 + a_2 + \cdots + a_n + b_1 + b_2 + \cdots + b_n}_{=\sum_{k=1}^n a_k + \sum_{k=1}^n b_k} \\ &= \underbrace{a_1 + b_1 + a_2 + b_2 + \cdots + a_n + b_n}_{=\sum_{k=1}^n (a_k + b_k)} \end{aligned}$$



Splitting, index shift and index reflection

Observation 3

Given two natural numbers $n, m \in \mathbb{N}$, we have:

$$\sum_{k=1}^{n+m} a_k = \sum_{k=1}^n a_k + \sum_{k=n+1}^{n+m} a_k \quad (\text{splitting})$$

$$\sum_{k=1}^n a_k = \sum_{k=1+m}^{n+m} a_{k-m} = \sum_{k=1-m}^{n-m} a_{k+m} \quad (\text{index shift})$$

$$\sum_{k=1}^n a_k = \sum_{k=1}^n a_{n-k+1} = \sum_{k=0}^{n-1} a_{n-k} \quad (\text{index reflection})$$

Proof

The three properties are simply different ways of writing and/or ordering the terms of the summations. □

Some important summations

Using the properties just seen, we now compute in closed form some important summations.

Sum of the first n natural numbers

Observation 4: sum of the first n natural numbers (without zero)

For every natural number $n \geq 1$, we have:

$$\sum_{k=1}^n k = \frac{n(n+1)}{2}$$

Proof

Using the index reflection:

$$\begin{aligned}\sum_{k=1}^n k &= \frac{1}{2} \left(\sum_{k=1}^n k + \sum_{k=1}^n k \right) = \frac{1}{2} \left(\sum_{k=1}^n k + \sum_{k=1}^n (n - k + 1) \right) \\ &= \frac{1}{2} \sum_{k=1}^n (k + n - k + 1) = \frac{1}{2} \sum_{k=1}^n (n + 1) = \frac{n(n+1)}{2}\end{aligned}$$



Sum of the first n odd numbers

Observation 5: sum of the first n odd numbers

For every natural number $n \geq 1$, we have:

$$\sum_{k=1}^n (2k - 1) = n^2$$

or equivalently

$$\sum_{k=0}^{n-1} (2k + 1) = n^2$$

Proof (1/2)

Using the properties of summations we have:

$$\begin{aligned}\sum_{k=1}^n (2k - 1) &= 2 \sum_{k=1}^n k - \sum_{k=1}^n 1 \\ &= 2 \left(\frac{n(n+1)}{2} \right) - n = n^2 + n - n = n^2\end{aligned}$$

Proof (2/2)

For the second form:

$$\begin{aligned}\sum_{k=0}^{n-1} (2k + 1) &= 2 \sum_{k=0}^{n-1} k + \sum_{k=0}^{n-1} 1 \\&= 2 \sum_{k=1}^n (k - 1) + n = 2 \left(\sum_{k=1}^n k - \sum_{k=1}^n 1 \right) + n \\&= 2 \left(\frac{n(n+1)}{2} - n \right) + n = n^2 + n - 2n + n = n^2\end{aligned}$$



Sum of the first n even numbers

Observation 6: sum of the first n even numbers (without zero)

For every natural number $n \geq 1$, we have:

$$\sum_{k=1}^n 2k = n(n+1)$$

Proof

$$\sum_{k=1}^n 2k = 2 \sum_{k=1}^n k = 2 \left(\frac{n(n+1)}{2} \right) = n(n+1)$$



Summary

Summations:

- ▶ $\sum_{k=1}^n a_k$ is a compact notation for $a_1 + a_2 + \cdots + a_n$; the index k is a **dummy index**
- ▶ Properties: product by a constant, constant term, sum of summations, splitting, index shift and index reflection

Important summations ($n \geq 1$):

$$\sum_{k=1}^n k = \frac{n(n+1)}{2}, \quad \sum_{k=1}^n (2k-1) = n^2, \quad \sum_{k=1}^n 2k = n(n+1)$$

Geometric progressions



Introduction: Geometric progressions

- ▶ In this section we introduce **geometric progressions**: sequences in which each term is obtained from the previous one by multiplying it by a constant, the **common ratio**
- ▶ We will derive the formula for the **sum of the first n terms** of a geometric progression
- ▶ This formula appears often in the course (for example in the study of geometric series)

Geometric progression

Definition 2: geometric progression

*A sequence of real numbers is a **geometric progression** if the ratio between each term (starting from the second one) and the previous one is constant. This constant is called the common ratio of the progression.*

Terms of a geometric progression

Given the first term $a \in \mathbb{R}$ and the common ratio $q \in \mathbb{R}$, the associated geometric progression is:

$$a, \quad a q, \quad a q^2, \quad a q^3, \quad a q^4, \quad \dots$$

Each term (starting from the second one) is obtained from the previous one by multiplying it by q . The k -th term ($k \in \mathbb{N}$, $k \geq 1$) can be written as $a q^{k-1}$ and we have:

$$a q^{1-1} = a q^0 = a \quad \text{first term, in position } k = 1$$

$$a q^{2-1} = a q^1 = a q \quad \text{second term, in position } k = 2$$

$$a q^{3-1} = a q^2 \quad \text{third term, in position } k = 3$$

$$a q^{4-1} = a q^3 \quad \text{fourth term, in position } k = 4$$

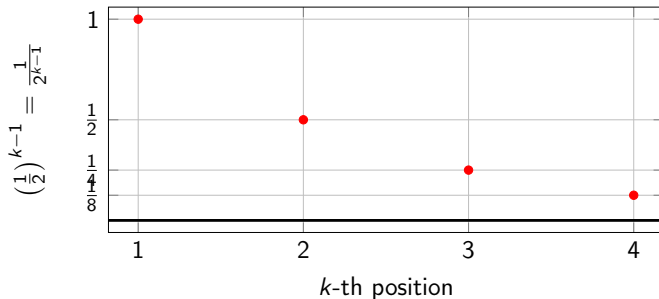
$\dots$

$\dots$

Example (1/2)

- With $a = 1$ and $q = \frac{1}{2}$, the first 4 terms are:

$$1, \quad \frac{1}{2}, \quad \frac{1}{4}, \quad \frac{1}{8}$$



Example (2/2)

- ▶ With $a = 1$ and $q = 2$, the first 4 terms are:

1, 2, 4, 8

Sums of the terms of geometric progressions

We now see how to compute in closed form the sum of the first n terms of a geometric progression.

Sum of the first n terms

Observation 7: sum of the first n terms of the geometric progression ($a = 1$)

Given $q \in \mathbb{R}_+$, for every natural number $n \geq 1$ we have:

$$\sum_{k=1}^n q^{k-1} = \begin{cases} \frac{q^n - 1}{q - 1} & \text{if } q \neq 1 \\ n & \text{otherwise} \end{cases}$$

Proof (1/2)

If $q \neq 1$, we prove the equation in the following equivalent form:

$$(q - 1) \sum_{k=1}^n q^{k-1} = q^n - 1$$

Applying the properties of summations, we get:

$$\begin{aligned}(q - 1) \sum_{k=1}^n q^{k-1} &= q \sum_{k=1}^n q^{k-1} - \sum_{k=1}^n q^{k-1} = \\&= \sum_{k=1}^n q^k - \sum_{k=1}^n q^{k-1} = \sum_{k=1}^n q^k - \sum_{k=0}^{n-1} q^k = \\&= \sum_{k=1}^{n-1} q^k + q^n - \left(1 + \sum_{k=1}^{n-1} q^k \right) = q^n - 1\end{aligned}$$

Proof (2/2)

If $q = 1$, we have instead:

$$\sum_{k=1}^n q^{k-1} = \sum_{k=1}^n 1 = n$$



Extension to the case $a \in \mathbb{R}$

- Given $q \in \mathbb{R}_+, n \in \mathbb{N}, n \geq 1$ and $a \in \mathbb{R}$, the formula extends as follows:

$$\sum_{k=1}^n a q^{k-1} = \begin{cases} a \left(\frac{q^n - 1}{q - 1} \right) & \text{if } q \neq 1 \\ a n & \text{otherwise} \end{cases}$$

since

$$\sum_{k=1}^n a q^{k-1} = a \sum_{k=1}^n q^{k-1}$$

Equivalent form

► Clearly:

$$\frac{q^n - 1}{q - 1} = \frac{1 - q^n}{1 - q}$$

and hence we also have:

$$\sum_{k=1}^n a q^{k-1} = \begin{cases} a \left(\frac{1-q^n}{1-q} \right) & \text{if } q \neq 1 \\ a n & \text{otherwise} \end{cases}$$

Example (1/2)

- With $a = 1$ and $q = \frac{1}{2}$, the first 4 terms are:

$$1, \frac{1}{2}, \frac{1}{4}, \frac{1}{8} \quad \text{and} \quad 1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} = \frac{8 + 4 + 2 + 1}{8} = \frac{15}{8}$$

the sum of the first $n = 4$ terms is given by the formula:

$$\sum_{k=1}^4 \left(\frac{1}{2}\right)^{k-1} = \sum_{k=1}^4 \frac{1}{2^{k-1}} = \frac{1 - \frac{1}{2^4}}{1 - \frac{1}{2}} = \frac{1 - \frac{1}{16}}{\frac{1}{2}} = \frac{15}{16} \cdot 2 = \frac{15}{8}$$

Example (2/2)

- With $a = 1$ and $q = 2$, the first 4 terms are:

$$1, \quad 2, \quad 4, \quad 8 \quad \text{and} \quad 1 + 2 + 4 + 8 = 15$$

the sum of the first $n = 4$ terms is given by the formula:

$$\sum_{k=1}^4 2^{k-1} = \frac{2^4 - 1}{2 - 1} = 16 - 1 = 15$$

Summary

Geometric progressions:

- ▶ First term a , common ratio q : terms $a, aq, aq^2, \dots$; the k -th term is aq^{k-1}

Sum of the first n terms:

- ▶ If $q \neq 1$:
$$\sum_{k=1}^n aq^{k-1} = a \frac{q^n - 1}{q - 1} = a \frac{1 - q^n}{1 - q}$$
- ▶ If $q = 1$:
$$\sum_{k=1}^n aq^{k-1} = an$$

Final summary

- ▶ The symbol $\sum$ allows us to write sums of many terms in compact form; its index is a dummy index
- ▶ The properties of summations (constants, sum, splitting, shift, reflection) allow us to compute sums in closed form
- ▶ Important summations:

$$\sum_{k=1}^n k = \frac{n(n+1)}{2}, \quad \sum_{k=1}^n (2k-1) = n^2, \quad \sum_{k=1}^n 2k = n(n+1)$$

- ▶ In a geometric progression with common ratio $q \neq 1$:

$$\sum_{k=1}^n a q^{k-1} = a \frac{q^n - 1}{q - 1}$$